

Introduction to Human Language Technology

Final Assignment

Period 1

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Section A

1.1 Train and Test dataset composition

Constructing a large training set consisted of WASA Tweets and MELD data set, aimed to provide the classifier with sufficient information that would boost its overall effectiveness and generalization capabilities. Indeed, upon combining the training data, they appeared to be ten times larger with a total of 21,157 instances compared to Eliza's 2,129. However, statistical analysis showed that both test and training data sets exhibit unequal distributions for most categories. In both, "neutral" emotion constitutes the dominant class, with 30 % in the training data and 39 % in the Eliza test data. Interestingly, the labels 'joy', 'surprise' and 'disgust' are unequally distributed in a similar way in the two databases. On the other hand, negative emotions like 'anger', 'fear' and 'sadness' are unequally distributed, adding an extra challenge on the future performance of the classifier, given that they are not only to a large extent context-dependent but also intrinsically difficult to distinguish from one another, resulting in potential overlaps of meaning.

1.2 Preprocessing and vectorization

The combined training data set was then transformed into an appropriate format for machine learning analysis, more specifically by using a combination of the BoW model and the TF-IDF weighting scheme. In an effort to remove noisy features, I set a threshold at 5, intending to exclude tokens that appear more than five times across the dataset, something that will ensure that terms belonging to the minority class 'disgust' will most likely be captured. Following this adjustment, Count Vectorizer generated a matrix representation with a vocabulary of 3436 unique word features representing each of the 21157 available words. It should be noted that I decided not to exclude stopwords, acknowledging their vital role in the context of sentiment

analysis, considering that the absence of words like ‘not’ can end up in the misclassification of an originally positive sentence into a negative one.

1.3 Expectations on the classifier’s performance

Given those characteristics, balancing the sets would certainly help provide a clearer idea regarding the classifier’s performance, however I decided to maintain the current distributions and observe the pitfalls during the error-analysis stage. In this context, I expect the classifier to overfit to ‘neutral’ and thus lead to a high recall score, as the overall data composition could help it correctly identify most of the neutral cases, but at the same time mislabel ambiguous cases as well, achieving a seemingly satisfactory accuracy. This will be probably happen at the expense of the underrepresented categories ‘disgust’ or ‘surprise’ and further complicate the distinction between negative emotions.

Section B

2.1 Discussing results

As it can be seen in the table, the classifier achieved an overall accuracy score of 56%, a mediocre performance which can be however justified by the fact that both the Eliza and the training data set were unbalanced and dominated by the neutral category (with 39% and 30% respectively). Thus, it is a logical consequence that even though the classifier identified 76% of neutral cases, only 57% of them are true positives. For instance, in the sentences “I’m crying right now” (Eliza: sadness / classifier: neutral) and "I get sick to my stomach when thinking about sauerkraut" (Eliza: disgust/ classifier: neutral), even though they actually contain strong, representative words for each emotion, they are labeled as neutral.

	precision	recall	f1-score	support
anger	0.4837398	0.4559387	0.4694280	261
disgust	0.4482759	0.1214953	0.1911765	107
fear	0.5333333	0.4965517	0.5142857	145
joy	0.5921450	0.5714286	0.5816024	343
neutral	0.5695187	0.7671068	0.6537084	833
sadness	0.6606335	0.4634921	0.5447761	315
surprise	0.2666667	0.0960000	0.1411765	125
accuracy			0.5622358	2129
macro avg	0.5077590	0.4245733	0.4423077	2129
weighted avg	0.5497899	0.5622358	0.5405488	2129

‘Joy’ appears to have the best results among the set with an F1 score of 60%, but again many instances have not been considered, even though joy was the second most prominent class in both sets, probably being affected by the overall problems traced in the distributional imbalances mentioned before. An interesting case is the sentence “I would be happy if you understood me” which was labeled in both cases as ‘joy’. This example shows that the model frequently assigns labels based on specific keywords, in this case the adjective ‘happy’, without being able to evaluate them depending on the context in which they appear. It should be noted that even though stop words have not been removed ‘if’ and ‘would’ do not help the classifier label the sentence with a more negative emotion label.

Regarding negative emotions, both ‘anger’ and ‘fear’ achieved recall scores near 50%. So, it can be assumed that approximately half of the actual cases were not detected, while for those detected there is again only 50% chance to be correct, with fear performing only slightly better. On the other hand, ‘sadness’ has a relatively high precision of 66%, but its low F1 score (54%) indicates that it fails to detect over half of the actual instances. At this point, the prediction made during the training stage is confirmed, as the classifier fails to capture the full range of negative emotions, potentially classifying them as neutrals if the emotion is not strongly expressed with an expected keyword or it is in case that it is highly context-bound. This is also confirmed by the confusion matrix, as anger, fear and sadness have 99, 45 and 108 cases mislabeled as ‘neutral’ respectively. For example, the following two cases, despite containing the words ‘scared’ and ‘mad’ they are mislabeled as neutral. ("Because I am scared that the world will end" (Eliza: fear/classifier: neutral), "It makes me mad!!" (Eliza: anger/classifier: neutral)). It must be noted that the Eliza dataset is a conversation, so the unavoidable exclusion of Eliza’s prompts might in certain cases deprive the speaker utterances from significant contextual information that justify the annotated labeling of the negative emotion. Finally, as expected, the least represented emotions ‘disgust’ and ‘surprise’ had the lowest performance metrics, being hardly traceable after the classification. This can be seen in the utterance "I get sick to my stomach when thinking about sauerkraut" (Eliza: disgust/ classifier: neutral), where the classifier cannot recognize that the whole expression ‘get sick to my stomach’ shows repulsion and disgust.

2.2 Conclusion

Overall, as anticipated during the training stage and its respective difficulties, the csv classifier overfitted to the neutral labelling as its high recall score and the confusion matrix demonstrate. More specifically, according to the matrix, ‘anger’, ‘sadness’ and ‘disgust’ were almost 50% likely to be mislabeled as neutral, while joy in roughly a third of its instances, justifying its relatively better performance. Also, it indeed failed to capture the nuances of the challenging negative categories and particularly struggled with the least represented categories of ‘surprise’ and ‘disgust’. However, it was expected that more cases of negative emotions would overlap with each other. The matrix shows that in 17 certain cases anger overlaps with fear and sadness, also fear is frequently misidentified as anger and sadness 17 times. The ambiguous utterances “I hate my family” (Eliza: anger/classifier: sadness) and "The possibility to fail in life is crippling me" (" Eliza: fear/classifier: anger) demonstrate this overlapping, but in the case of the first sentence can be logically justified.

Apparently, a vast majority of negative emotions were more vulnerable towards being neutralized rather than being affected by those hardly distinguishable nuances. To conclude, despite the fact that the model’s overall accuracy is approaching 60%, the low precision (50%), recall (42%) and F1 score (44%) when treating each class equally show a problematic treatment of less represented emotions compared to more represented ones, highlighting the need for further improvements.

SECTION C

3.1 Preprocessing of GO-classifier and expectations

Another approach involved training and deploying the GO classifier on the Eliza annotated dataset, in an effort to examine the potential effects of using a classifier that is based on GoEmotions, a carefully curated dataset for fine-grained emotion prediction. Mapping the GO-classifier to Ekman’s seven basic universal categories is expected to help in better generalizing the emotional content of Eliza’s dataset. Also, the three sentiment values ‘positive’, ‘negative’ and ‘ambiguous’ are expected to accurately predict certain ambiguous cases. Recognizing the unequal distribution of the Eliza dataset and more specifically the predominance of the ‘neutral’

emotion as a dynamic majority class that comprises almost 40% of the total, I initially attempted to balance the seven categories. Nevertheless, this adjustment did not improve the overall performance. So, I decided to maintain the original unequal distribution, baring in mind the potential biases this may create in the statistical analysis. I then proceeded with removing unusual tokens as well as stopwords, since after testing, it seemed that those common tokens, despite being generally significant in sentiment analysis, created noise. Also, a threshold of 0.05 was set for the mapping, aiming at taking into account only significant emotion scores and again minimizing noise. After these preprocessing steps, the GO classifier was applied to each cleaned speaker utterance, showing both mapped emotion scores and averaged emotion scores. Overall, it is expected that the model might become biased towards the majority class, by potentially classifying ambiguous cases as ‘neutral’. This could unavoidably result in a decrease in recall for minority classes. It should also be mentioned that because of the fact that GO-emotions are more finely-tuned and trained on larger and diverse datasets, they will most probably recognize nuances that the Eliza annotations cannot by default represent.

3.2 Results

The GO classifier achieved an overall accuracy of 63.22%. It can be thus inferred that it performed better than the SCV classifier, having a better precision and recall across all classes. It however exhibits certain common patterns which stem from the unbalanced distribution of the Eliza data set and the subsequent prioritization of the ‘neutral’ category.

Classification Report:					
	precision	recall	f1-score	support	
anger	0.7222	0.3985	0.5136	261	
disgust	0.7045	0.2897	0.4106	107	
fear	0.8750	0.5310	0.6609	145	
joy	0.6185	0.7230	0.6667	343	
neutral	0.5740	0.8283	0.6781	833	
sadness	0.8485	0.4444	0.5833	315	
surprise	0.6588	0.4480	0.5333	125	
accuracy			0.6322	2129	
macro avg	0.7145	0.5233	0.5781	2129	
weighted avg	0.6720	0.6322	0.6190	2129	

Once again, as predicted, the classifier shows a tendency to overfit the neutral emotion label. So, the neutral category with a recall score of almost 83% contains the vast majority of misclassifications and at the same time according to the confusion matrix has the most true positives (690). Given the particularities of the data set, 'joy' also performed relatively well, with an F1 score of 67% ,but more than a quarter of its instances was labeled as neutral, showcasing the aforementioned overfitting issue. In the case of the sentence "I would be happy if you understood me", GO- classifier is not expected to correctly label it with a negative label, as stop words have been removed, and thus correctly identifies 'happy' as 'joy'.

Regarding the negative class of labels, there are small but noticeable improvements for the precision values of 'anger', 'fear' and 'sadness' while the recall also seems to be better. It should be thus noted that while 'neutral' label still overlaps with other emotions, particularly in the case of anger and sadness, the differentiation between those often ambiguous utterance cases appears to be handled better than the SCV classifier. The following example that was previously labeled as neutral by SCV, now exhibits a higher score for 'sadness' illustrating this enhanced performance ('sadness': [0.0610, 0.7441], 'neutral': [0.054]). The same can be said for the previously neutral-labeled example "I'm so scared that the world will end". Now 'fear' is the most probable predicted label ('fear': [0.7435], 'neutral': [0.05671]). However, the previously analyzed example "It makes me mad!" is again labeled as neutral because despite 'anger's' good score, the latter is facing intense competition by the majority class ('anger': [0.1796, 0.1532], 'surprise': [0.0554], 'neutral': [0.3669])

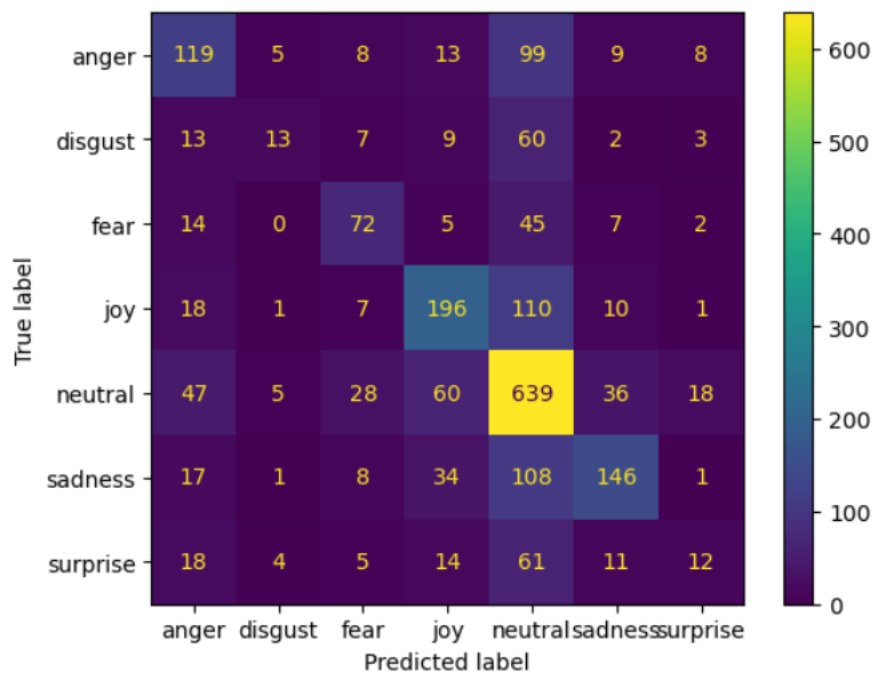
Concerning the least represented emotions 'surprise' and 'disgust' almost half of the cases are mismatches, which can be justified by their small representation in the Eliza dataset. A worth mentioning positive example is the sentence "Disgusted at myself" which is this time labelled correctly as 'disgust' ('anger': [0.0517], 'disgust': [0.6964]) compared to CSV classifier's prediction, according to which it expresses a neutral sentiment. Overall, comparing the two statistical analyses, as indicated by the GO-classifier's macro average F1 score of 57%, the model's performance is indeed slightly more balanced across the various classes.

3.3 Improvements

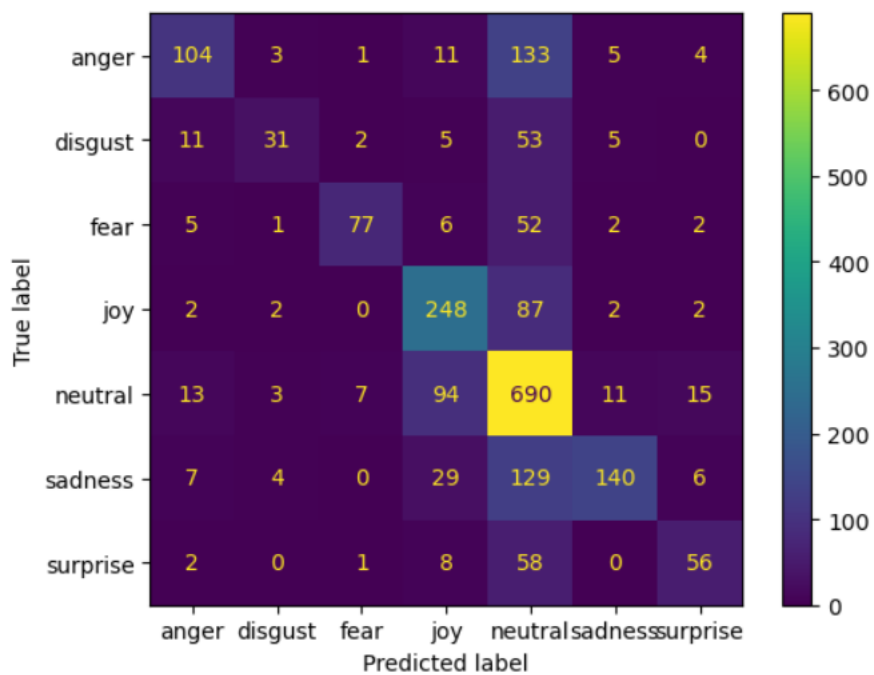
To sum up, both classifiers could significantly benefit from balancing training and test data, by for instance reducing the number of instances for the majority classes, or even synthetically expanding the data set to minimize the risk of missing important information. Techniques that also manage to take context into consideration could drastically reduce unwanted mislabeling, as well as provide the opportunity to better distinguish between nuances among the subcategories of positive and negative emotions.

Appendices

Confusion matrix for SCV classifier



Confusion matrix for GO-classifier



Distribution of data from Eliza dataset

The labels are: ['anger', 'disgust', 'fear', 'joy', 'neutral', 'sadness', 'surprise']

The labels and their respective values in the training set are: OrderedDict([('anger', 261), ('disgust', 107), ('fear', 145), ('joy', 343), ('neutral', 833), ('sadness', 315), ('surprise', 125)])

Total of values 2129

